

AI Veterinary Assistance: Enhancing Clinical Decision-making in Animal Healthcare

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Project Guide:

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Project Domain:

Artificial Intelligence

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Introduction & Project Objectives

Introduction

With rising pet ownership, veterinarians face higher workloads. This can affect efficiency in decision-making. Our project, AI Veterinary Assistance (AVA), leverages a Large Language Model (LLM) to support veterinarians during consultations.

Objectives

- Develop AVA framework to assist consultations.
- Extract symptoms from unstructured text.
- Predict probable diseases using a certified database.
- Recommend follow-up questions for diagnosis.
- Improve efficiency and reliability of consultations.

Purpose and Need

Purpose

To design a reliable AI assistant that integrates with veterinary workflows to enhance decision-making.

Need

- **Support:** Assist, not replace, veterinarians.
- **Structure:** Convert unstructured input into clinical data.
- **Transparency:** Provide explainable reasoning.
- **Scalability:** Easy updates for new diseases and symptoms.

Scope of the Project

- Real-time analysis of consultation records.
- Symptom extraction using GPT-based LLM.
- Match symptoms with a certified disease database.
- Predict Top-K probable diseases.
- Generate follow-up questions for veterinarians.
- Focused on companion animals (dogs, cats).

- ➊ Input: Consultation text from veterinarian–owner interaction.
- ➋ Symptom Extraction using LLM.
- ➌ Cross-check with certified symptom–disease database.
- ➍ Predict top probable diseases.
- ➎ Recommend relevant follow-up questions.
- ➏ Provide structured, transparent output to veterinarian.

Base Paper: AI Veterinary Assistance (AVA)

Title

AI Veterinary Assistance: Enhancing Clinical Decision-making in Animal Healthcare (2025, IEEE Access)

Objectives

- Propose AVA, an LLM-based framework to assist in veterinary consultations.
- Automatically extract symptoms from consultation records using GPT-4o.
- Predict diseases and recommend questions by matching symptoms to a certified disease-symptom database.
- Evaluate AVA on both generated and real-world consultation records to demonstrate its effectiveness.

Base Paper: System Architecture



Figure: Overview of the AVA framework. Uses an LLM for symptom extraction, then accesses a database for disease prediction and question recommendation.

Base Paper: Design Methodology & Implementation

Design Methodology

- **Two-Stage Process:** 1. Symptom extraction via LLM (GPT-4o). 2. Database matching for disease prediction and question generation.
- **Database-Driven:** Uses a veterinarian-certified database of 244 diseases and 464 symptoms for explainability and easy updates.
- **In-Context Learning (ICL):** Prompts with few-shot examples for symptom extraction, avoiding fine-tuning.

Implementation Strategy

- System takes raw consultation text as input.
- Detailed prompt, including an official symptom list, is sent to GPT-4o API.
- Extracted symptoms query the local database.
- Diseases are ranked by matched symptom frequency.
- Follow-up questions generated from the most frequent or decisive

Base Paper: Results & Future Scope

Results

- **Generated Data:** Top-3 accuracy 91.4%, Top-5 93.4%, Top-10 95.9%; Symptom extraction accuracy 79.9%.
- **Real-World Data:** Top-3 38.6%, Top-10 51.5%, outperforming baselines despite messy data.

Future Scope

- Improve performance on unstructured, real-world data.
- Enhance disease-symptom database with granular info (e.g., symptom severity).
- Develop a mechanism for user correction of symptom extraction errors.
- Validate question recommendation strategies with practicing veterinarians.

Title

VetLLM: Large Language Model for Predicting Diagnosis from Veterinary Notes (2024, Pacific Symposium on Biocomputing)

Objectives

- Evaluate the effectiveness of fine-tuned open-source LLMs for multi-label diagnosis coding from free-text clinical notes.
- Compare a fine-tuned model (VetLLM) against zero-shot LLMs and other baselines.
- Study the data efficiency of LoRA (Low-Rank Adaptation) fine-tuning with small subsets of veterinary notes.

Architecture

- **Base Model:** Alpaca-7B, an instruction-tuned LLM.
- **Fine-Tuning:** VetLLM is created by fine-tuning Alpaca-7B using LoRA on a small dataset of veterinary notes (5k).
- **Input:** Free-text clinical notes from two veterinary sources (CSU and PP).
- **Task:** Multi-label diagnosis classification for nine prevalent diseases.
- **Prompting Strategy:** A "one-disease-at-a-time" prompt is used, where the model is asked a yes/no question for each potential disease to simplify output parsing.

System Architecture

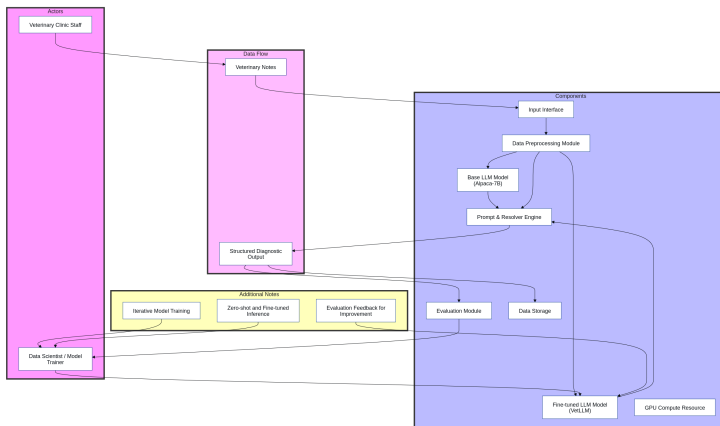


Figure: Overview of the system architecture.

Paper 1: Design Methodology & Implementation

Design Methodology

- Leverage LLMs, fine-tuned for veterinary domain using LoRA for efficiency.
- Create a diagnostic tool with minimal computational cost.
- Achieve strong performance with a small, domain-specific dataset.

Implementation Strategy

- Fine-tune Alpaca-7B on 5,000 notes from CSU dataset.
- Evaluate on CSU test set and out-of-domain PP dataset.
- Measure performance using Macro F1 and Exact Match Ratio.

Paper 1: Results & Future Scope

Results

- VetLLM (fine-tuned) significantly outperformed the zero-shot Alpaca-7B model.
- On the out-of-domain PP dataset, VetLLM achieved a Macro F1 of 0.637, a substantial improvement over the zero-shot model's 0.389.
- Showed high data efficiency, surpassing the VetTag baseline with as few as 200 fine-tuning notes.

Future Scope

- Address model sensitivity to prompt formatting.
- Develop multi-label prompts to reduce the need for multiple inference passes.
- Broader external validation on more diverse datasets.
- Combine structured coding (VetLLM) with reasoning capabilities (like AVA) for a more comprehensive tool.

Title

When Pigs Get Sick: Multi-Agent AI for Swine Disease Detection (2025, arXiv preprint)

Objectives

- Address lack of veterinary expertise in swine disease detection in remote areas.
- Propose a multi-agent AI system using RAG for accurate diagnosis and treatment.
- Improve food security by supporting sustainable livestock management.

Paper 2: System Architecture

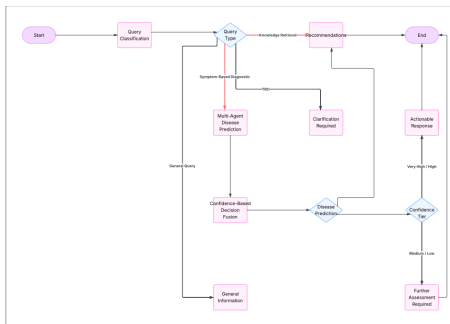


Figure: Diagnostic conversation flow showing the multi-agent system in action.

- **Query Classifier:** Routes input to correct agent (e.g., diagnosis vs. general info).
- **Multi-Agent Disease Prediction:** Specialized agents collaborate, combining predictions using confidence-weighted fusion.
- **Recommendations Module:** RAG grounds LLM-generated advice in reliable documents.

Paper 2: Design Methodology & Implementation

Design Methodology

- Conversational, multi-turn approach for systematic symptom gathering (General → External → Specific).
- Multi-agent architecture with "expert" agents for improved robustness.
- RAG ensures factual recommendations, reducing hallucination risks.

Implementation Strategy

- Built and tested with multiple LLMs (01-mini, GPT-4o, Gemini-1.5-Pro).
- Evaluated on 812 expert-created questions covering classification, diagnosis, and knowledge retrieval.

Paper 2: Results & Future Scope

Results

- Achieved 95.23% accuracy in query classification.
- GPT-4o was the fastest and most accurate in disease diagnosis (90.63%).
- RAG-based knowledge retrieval outperformed baseline with more relevant answers.

Future Scope

- Integrate multimodal inputs (e.g., lesion images, cough audio) for better diagnosis.
- Enhance context-aware query disambiguation.
- Expand the knowledge base for emerging and region-specific swine diseases.

Title

Toward Expert-Level Medical Question Answering with LLMs (Review of Med-PaLM 2) (2025, Nature Medicine)

Objectives

- Develop Med-PaLM 2, an expert-level medical question-answering system.
- Fine-tune a powerful base LLM (PaLM 2) for medical-specific tasks.
- Ensure safe, accurate, and helpful long-form answers to medical queries.
- Validate performance against medical exam benchmarks and evaluations by physicians.

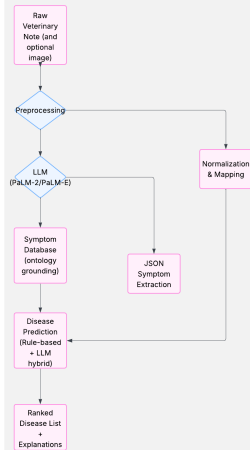
Three Pillars of Med-PaLM 2

- **Improved Base LLM:** Built on PaLM 2 with better scaling and training data.
- **Medical Fine-Tuning:** Aligned to the medical domain with MultiMedQA datasets (medical exams, research, consumer questions).
- **Advanced Prompting Strategies:** Includes "ensemble refinement" and "chain-of-retrieval" to improve reasoning and ground answers.

System Architecture

Veterinary IE Pipeline System Architecture

Veterinary IE Pipeline System Architecture



Paper 3: Design Methodology & Implementation

Design Methodology

- Combines the power of a general LLM with medical domain fine-tuning for expert-level performance.
- Focuses on safety and accuracy through advanced prompting techniques.
- Rigorous evaluation using both quantitative benchmarks and qualitative ratings from clinicians.

Implementation Strategy

- Fine-tuned on the MultiMedQA dataset.
- Evaluated on MedQA (USMLE-style questions) and adversarial datasets for safety and bias.
- Physician panel rated answers across 9 clinical axes (e.g., accuracy, harm, bias).

Paper 3: Results & Future Scope

Results

- Achieved 86.5% accuracy on the MedQA dataset, improving by over 19% compared to the first Med-PaLM.
- Physicians preferred Med-PaLM 2's answers over others in 8 out of 9 clinical axes.
- Responses were rated as safe as those from human physicians.

Future Scope

- Extend the model for multi-turn dialogues and clinical workflows.
- Improve safety and bias evaluation methods.
- Integrate with external databases and search tools for enhanced reliability—relevant to our AVA project.

Title

The Potential Application of Artificial Intelligence in Veterinary Clinical Practice and Biomedical Research (2025)

Objectives

- Provide an overview of AI applications in veterinary medicine.
- Explore how AI can transform diagnostics, treatment, epidemiology, and research.
- Identify limitations and challenges to AI adoption in veterinary science.

Paper 4: System Architecture (Application Areas)

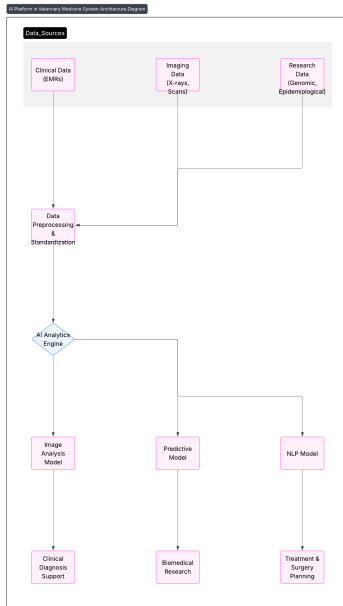
AI in Clinical Practice

- **Diagnosis & Monitoring:** Early disease detection through imaging (e.g., texture analysis of brain scans) and data analysis.
- **Epidemiology:** Predicting infection risks and forecasting zoonotic outbreaks.
- **Treatment & Surgery:** Remote health monitoring, personalized treatments to reduce AMR, and enhancing surgical precision (e.g., STAR robot).

AI in Biomedical Research

- **Drug Discovery:** Identifying new drug targets and generating novel antibiotic candidates quickly.
- **Cancer Research:** Improving cancer detection (e.g., Sybil model for lung cancer risk).

System Architecture



Paper 4: Design Methodology & Implementation

Design Methodology

As a review, the paper surveys various AI techniques applied in veterinary science, including:

- **Machine Learning for Imaging:** Using CNNs and texture analysis to interpret medical images.
- **Natural Language Processing (NLP):** Predicting virus host ranges using models like Word2vec.
- **Predictive Modeling:** Forecasting disease outbreaks and optimal breeding times with algorithms.

Implementation Strategy

Successful implementation requires:

- Collaboration between veterinarians and AI experts.
- Development of large, standardized, and structured datasets.
- Clear regulatory guidelines for AI use in clinical settings.

Paper 4: Results (Findings) & Future Scope

Results / Key Findings

- AI shows significant potential to improve accuracy and efficiency in veterinary medicine.
- Challenges are not just technological but also related to data availability, quality, and regulation.
- The lack of large, structured datasets is a major bottleneck for developing robust AI models.

Future Scope

- Utilize NLP and Large Language Models for analyzing clinical notes (directly relevant to AVA).
- Develop standardized, shared databases to facilitate large-scale research.
- Combine AI tools with traditional veterinary methods for a hybrid approach to healthcare.

Literature Review: Result Comparison

Paper	Key Results	Method/Tech	Performance
Base (AVA)	High accuracy on generated data; recommends follow-up questions	GPT-4o + disease-symptom DB	Gen: 95.9% Top-10, Real: 51.5% Top-10
Paper 1 (VetLLM)	Fine-tuned LLM improves diagnosis; data-efficient	Alpaca-7B + LoRA	Macro F1: 0.747 / 0.637
Paper 2 (Swine AI)	High swine query classification	Multi-agent + RAG	90.63% diagnosis; better retrieval
Paper 3 (Med-PaLM 2)	Expert-level medical Q&A	Med-fine-tuned PaLM 2	MedQA: 86.5%
Paper 4 (AI Review)	Survey: AI potential limited by data	ML, NLP, predictive	N/A; standardized datasets needed

Expected Outcomes

- Faster and more reliable consultations.
- Structured clinical records from raw text.
- AI-driven support without replacing veterinarians.
- Transparent reasoning for predictions.
- Improved trust in AI adoption for veterinary healthcare.

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